

**YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 2: CHEMICAL CALCULATIONS**

NAME: _____

Total: $\frac{\quad}{34}$

1. Calculate the formula mass for the following substances.

NAME	FORMULA	FORMULA MASS (g mol^{-1})
sodium phosphate		
iron (III) nitrate		
calcium hydrogencarbonate		
barium sulphide		
zinc acetate		

(10)

2. Convert the following to moles or mass as necessary.

(a) 36.2 g of copper (II) nitrate.

(b) 2.34 moles of aluminium hydroxide.

(b) 1.34 kg of iron (III) oxide.

(d) 1.15×10^3 moles of sulphur trioxide.

(8)

3. 27.2 g of aluminium carbonate is reacted with excess sulphuric acid. The equation for the reaction is as follows.



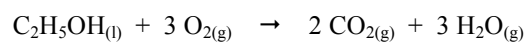
- (a) Calculate the mass of sulphuric acid used in the reaction?

(5)

- (b) What mass of carbon dioxide would be produced?

(3)

4. Ethanol burns in air according to the following equation.



If 3.85×10^2 g of ethanol burns, calculate:

- (a) the mass of O_2 required.

(5)

- (b) the mass of CO_2 produced.

(3)